香港科技大學(廣州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

Quantum Cellular Automata Simulation in Multi-species Rydberg Atom Array

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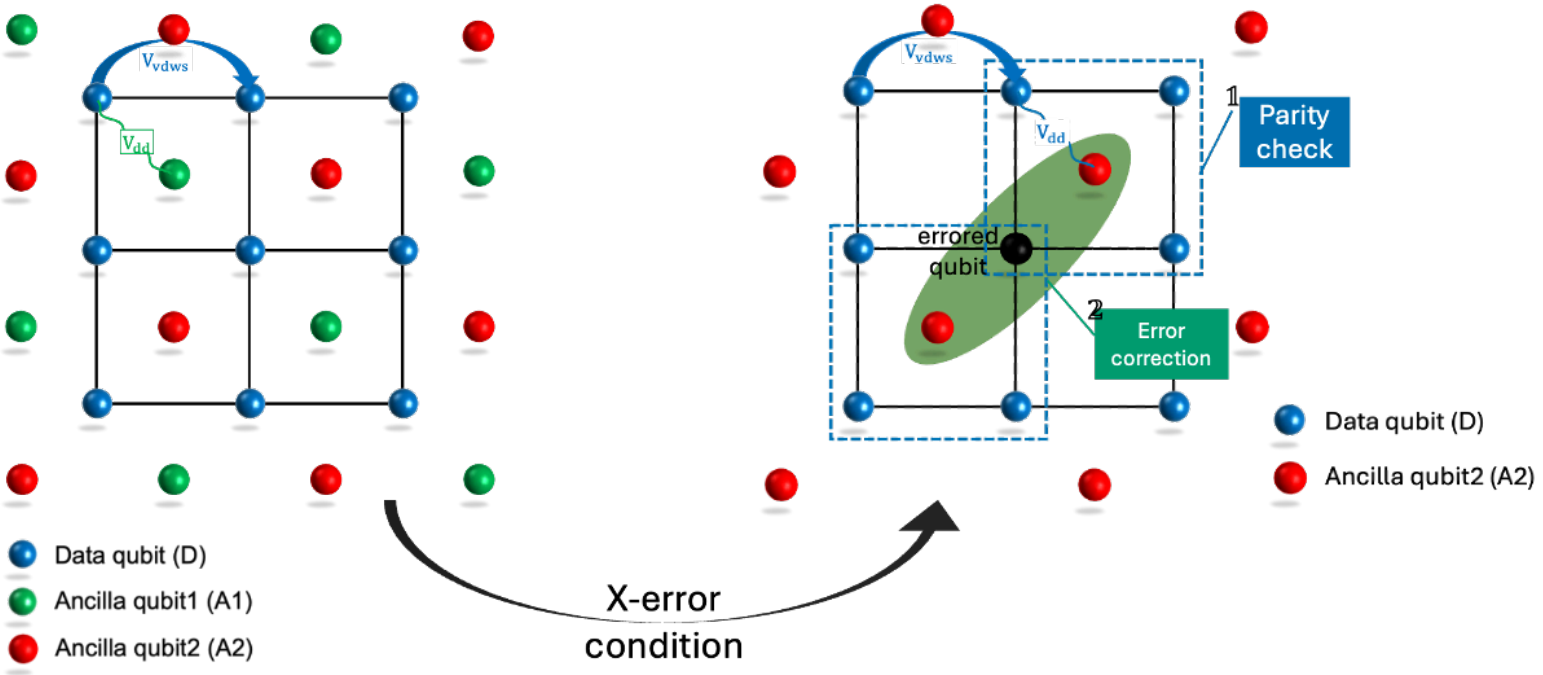
Theoretical model

- Multi-species array



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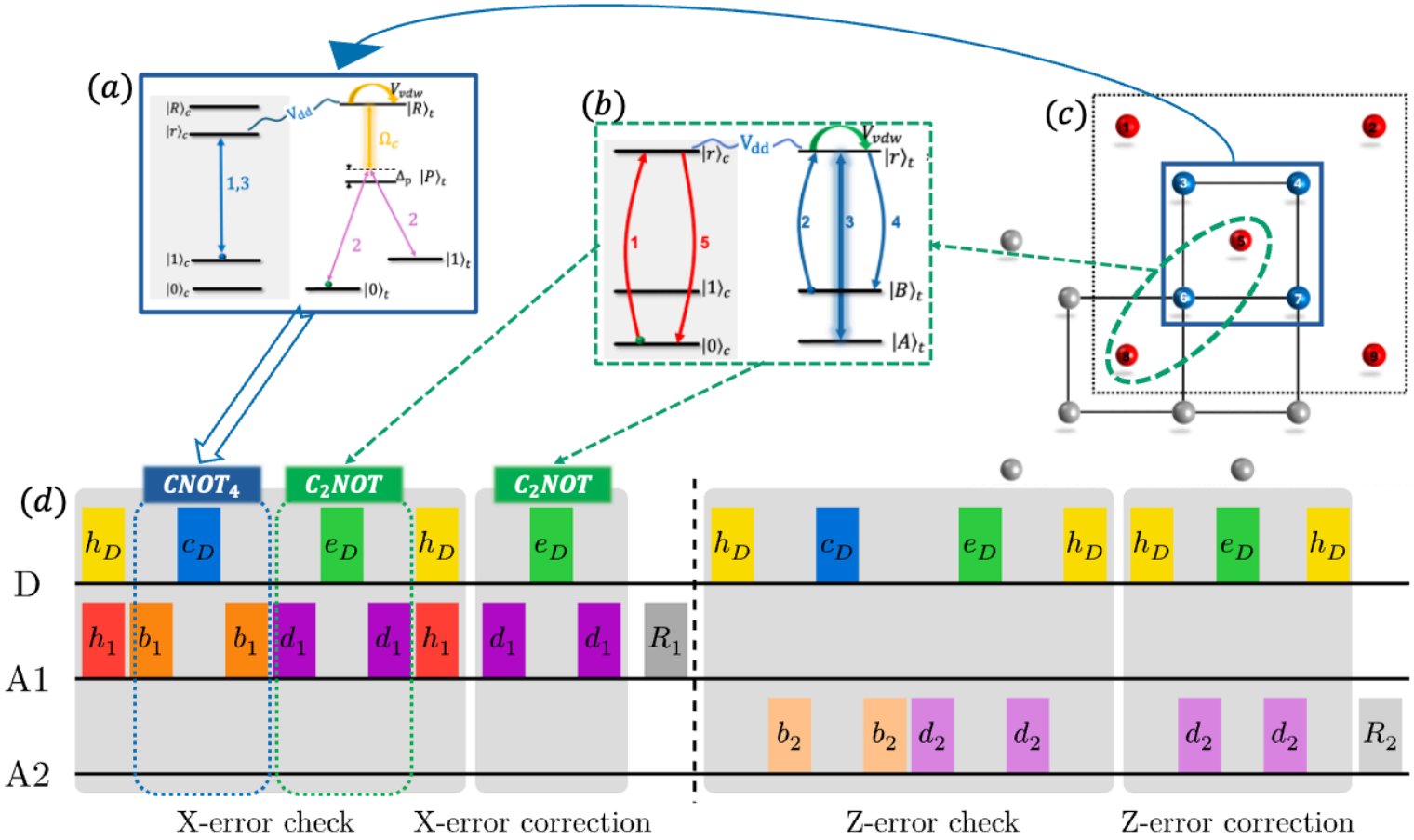
Global pulse sequence and level scheme



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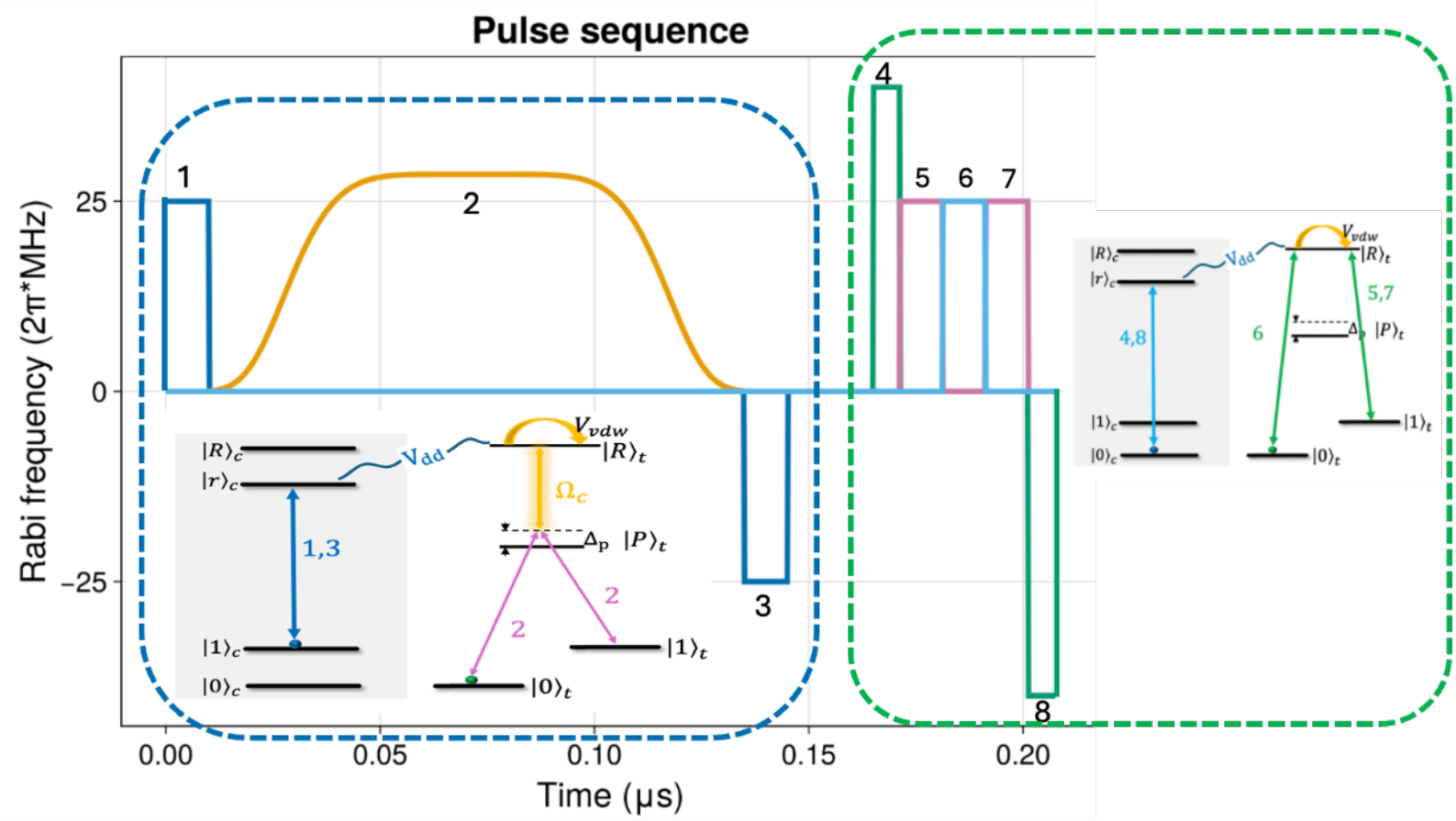
Pulse shape



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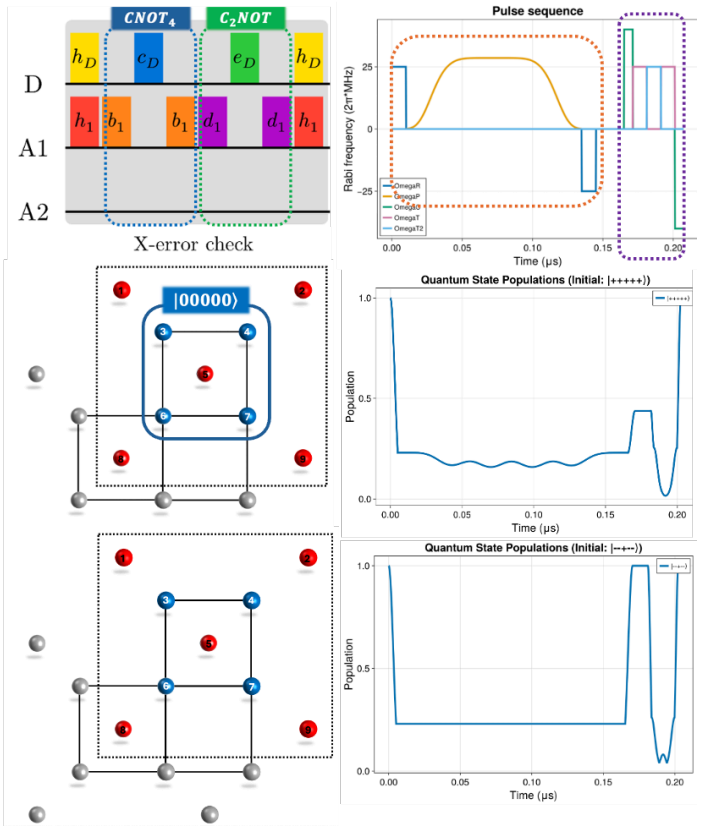
Pulse-level simulaiton

- No flipped error



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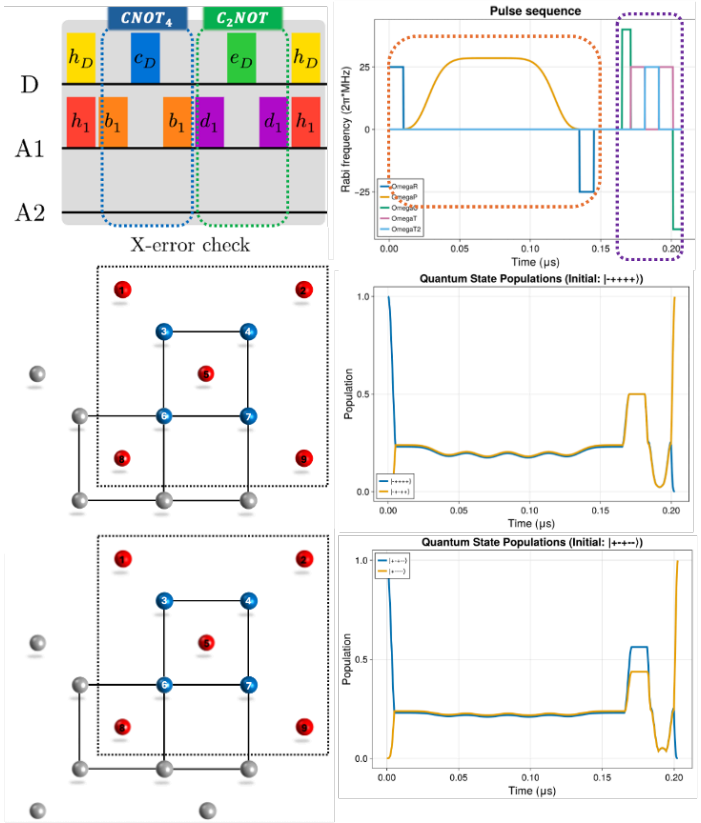
Pulse-level simulaiton

- Single X-error



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cellular automata

What is CA (cellular automata)

- Defination: CA consists of an array of identical cells, each of which can be in one of a finite number of states. The array evolves in discrete time steps according to a local, identical interaction rule.[1], [2], [3]
- Cell state: each cell can take one of a finite number of possible states.
- Evolution: The whole array evolves in discrete time steps by iterating a function G, known as the global evolution.

cellular automata

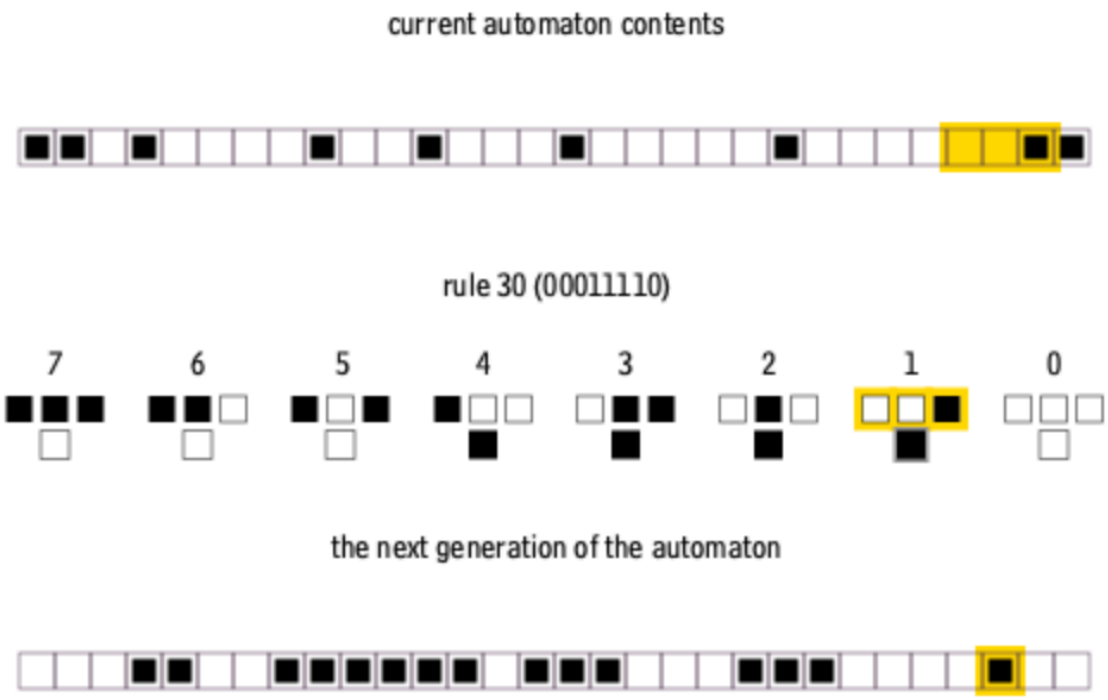

Figure 6: Cellular Automata

cellular automata

CA (cellular automata) for universal computation

Universal CA is universal for classical computation [4], which, likes Turing machine, exhibits universality.

Example: Game of Life (Universal for TM)

CA have been widely used in the study of complex systems, such as the Game of Life by Conway.

In 2D grid, each cell has 8 neighbors.

If the cell is alive, then it stays alive if it has either 2 or 3 live neighbors

If the cell is dead, then it springs to life only in the case that it has 3 live neighbors

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cellular automata

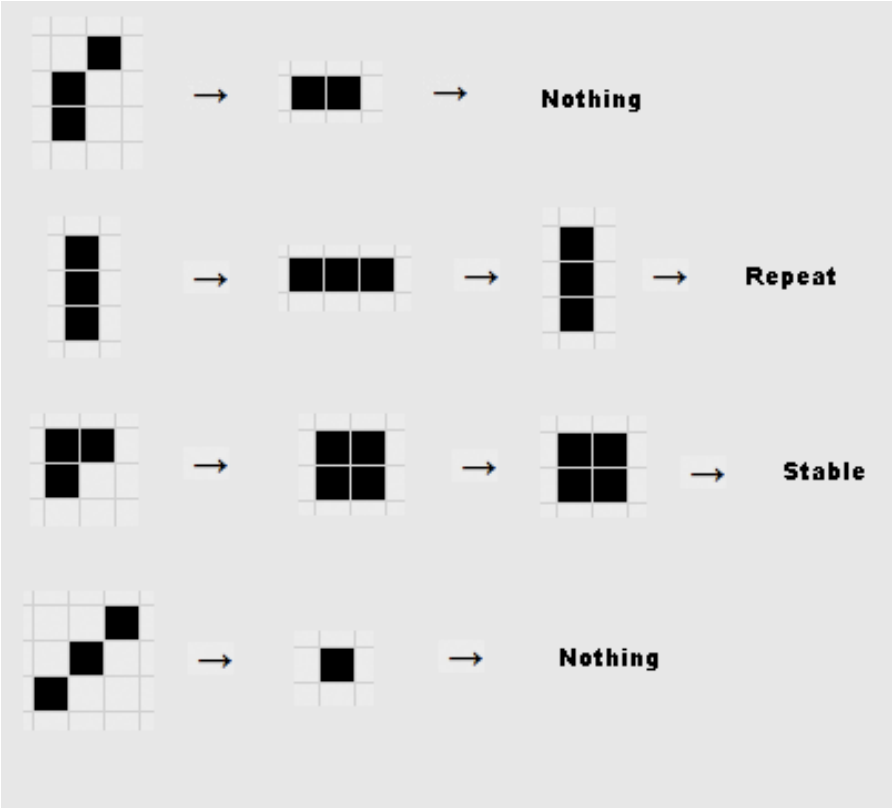

Figure 7: Game of Life by Conway

Quantum Cellular Automata

QCA[5], [6]

QCA is a quantum version of CA, which can perform quantum computation.

- cells → qubits
- classical states → quantum states(superposition)
- local classical function → unitary operator[7]

Characteristics:

- no external control (Low noise)
- Evolution: evolve unitarily in discrete time steps

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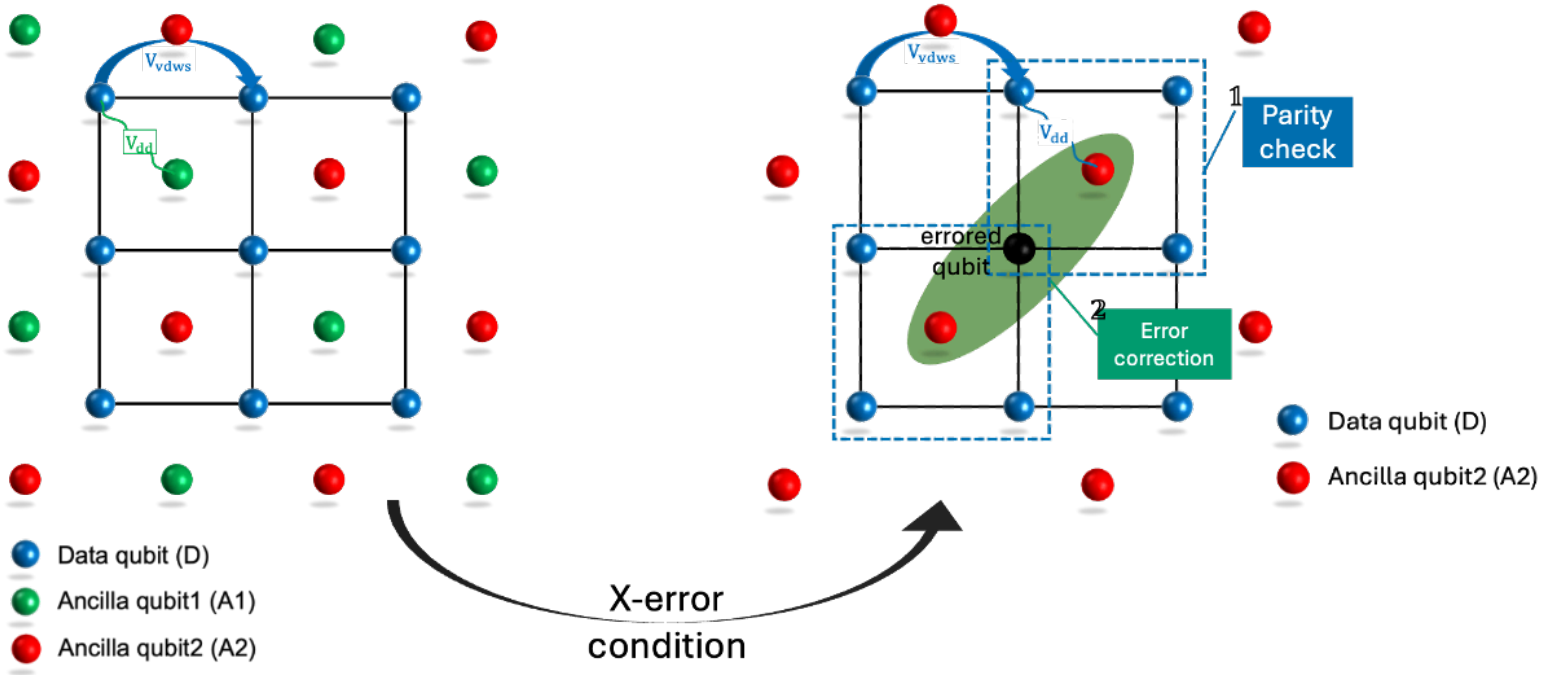
QCA simulation in our system

- Multi-species array



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QCA simulation in our system

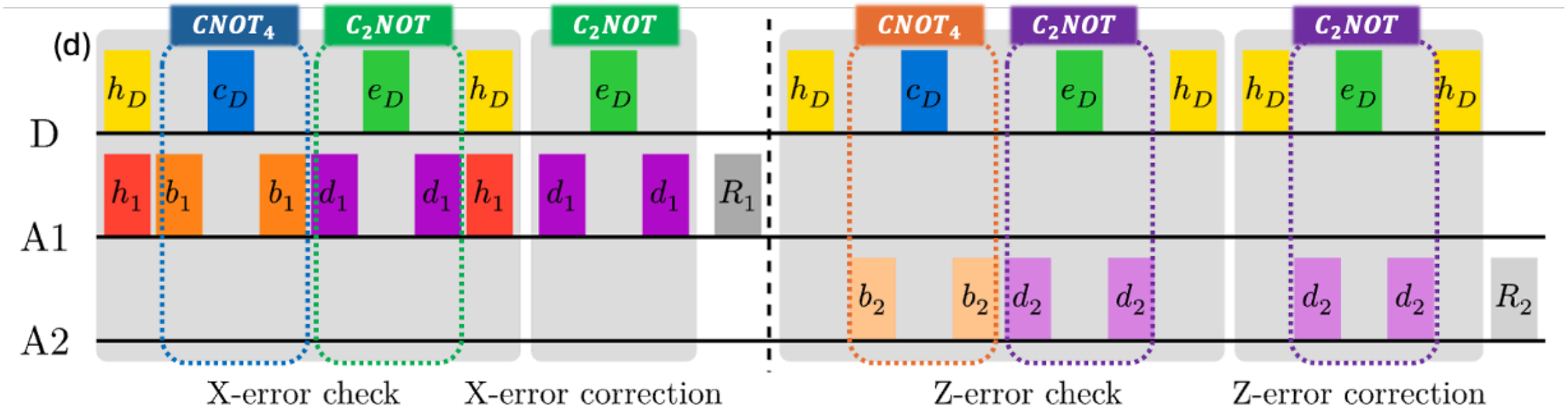


Figure 9: QCA simulation in our system

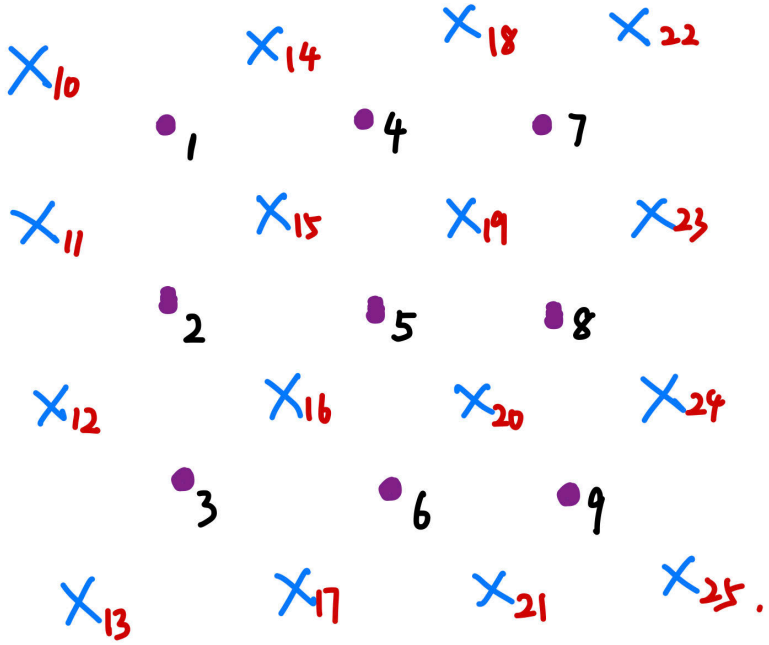
QCA simulation in our system

Effect of different pulses

Pulse	**Action**	**Effect**
$h_{D/1/2}$	Hadamard on D/A1/A2	$H\ \lvert\psi\rangle$
$b_{1/2}$	Excite A1/A2	$\lvert 1\rangle \leftrightarrow \lvert r\rangle$
c_D (EIT)	Conditional flip on D	Neighbor in $\lvert r\rangle$: $\lvert 0\rangle \leftrightarrow \lvert 1\rangle$
$d_{1/2}$	Excite A1/A2	$\lvert 0\rangle \leftrightarrow \lvert r\rangle$
e_D	Conditional flip on D	No neighbor in $\lvert r\rangle$: $\lvert 0\rangle \leftrightarrow \lvert 1\rangle$

Lattice construction

purple points means data qubits and blue “x” means ancilla qubits.



Local rules

Step	Operation	Description
1	Hadamard Rotation	Apply H gate to all qubits: $\|\psi\rangle \to H^{\otimes 32} \|\psi\rangle$
2	π Pulse (Ancilla)	Apply X gate to ancilla qubits for state flip $\|1\rangle \leftrightarrow \|r\rangle$ ($A = \{11, 13, 14, 16, 19, 21, 22, 24\}$): $\|\psi\rangle \to X_A \|\psi\rangle$.
3	Conditional EIT Pulse	Apply conditional flip c_D on data qubits: If $\exists$ neighbor in $\|r\rangle$: $\|0\rangle \leftrightarrow \|1\rangle$
4	π Pulse (Repeat)	Repeat Step 2: $\|\psi\rangle \to X_A \|\psi\rangle$
5	C2NOT Operation	Multi-controlled NOT on data qubits with ancilla controls: $\|\psi\rangle \to \mathrm{C2NOT}_{A\to D} \|\psi\rangle$
6	Hadamard Rotation	Apply H gate to all qubits: $\|\psi\rangle \to H^{\otimes 32} \|\psi\rangle$
7	State Extraction	Measure computational basis: $\{\|0\rangle, \|1\rangle\}^{\otimes 32}$

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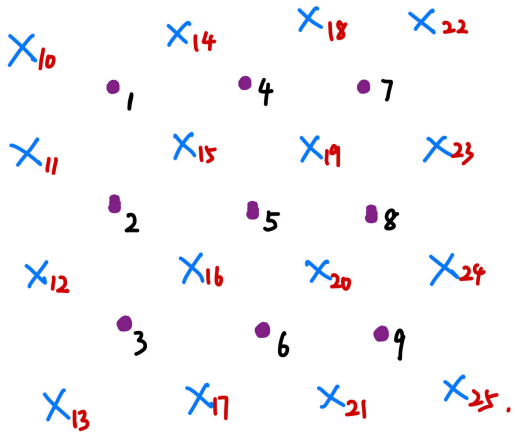
Simulation results

Test Case	Initial State	Error Config	Ancilla Response
No Error	$\|0\rangle^{\otimes 32}$	None	All remain $\|0\rangle$
Single X-Error	$\|0\rangle^{\otimes 32}$	X on qubit 5	Qubits 16, 19 flip to $\|1\rangle$
X-Error (All $\|1\rangle$)	Data: $\|1\rangle^{\otimes 16}$ Ancilla: $\|0\rangle^{\otimes 16}$	X on qubit 5	Qubits 13, 16, 19, 22 flip to $\|1\rangle$



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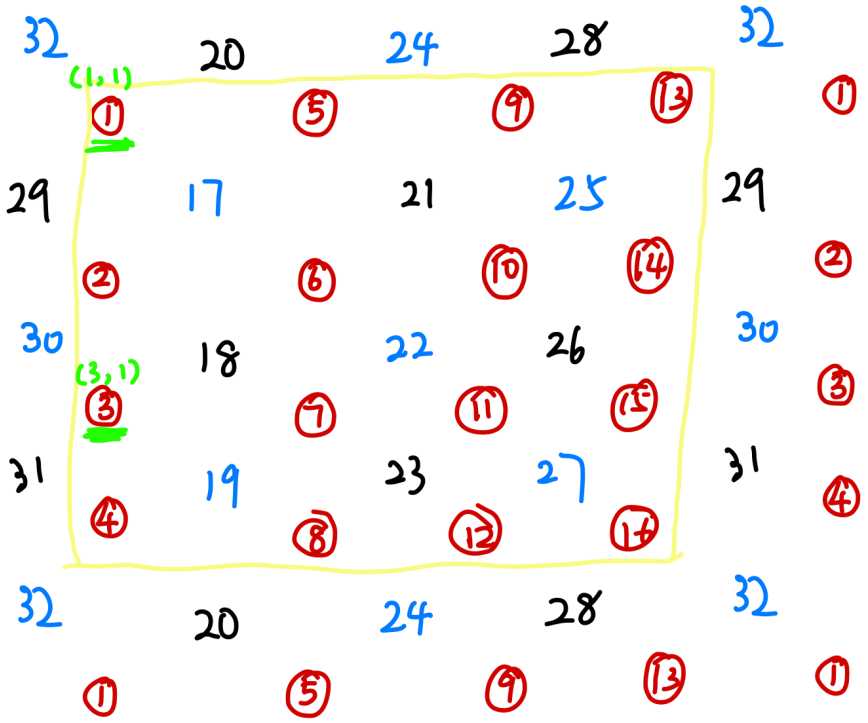
Periodic boundary condition:



Ancilla X-error (black)

Ancilla Z-error (blue)

Bibliography

- Periodic boundary condition: Requiring larger lattice to meet the condition.
- How to represent the state of the whole lattice with discrete steps of QCA evolution?

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[1] J. Kari, "On the Circuit Depth of Structurally Reversible Cellular Automata," *Fundamenta Informaticae*, vol. 38, no. 1–2, pp. 93–107, 1999, doi: 10.3233/FI-1999-381208.

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[7] D. Nagaj and P. Wocjan, "Hamiltonian quantum cellular automata in one dimension," *Physical Review A*, vol. 78, no. 3, Sep. 2008, doi: 10.1103/physreva.78.032311.